Degradation models for the ORB-SLAM3 correction pipeline

Handoff brief · 16 October 2026 · branch `experiments/denoising` · full detail in `reports/Degradation correction framework.md`

1. What this project is

We are building a **degradation-aware correction framework** around ORB-SLAM3 stereo-inertial (OASIS fork, Docker image `oasis-orbslam3:24.04` on brd-lambda). It has two halves: corrections applied to sensor data **before** SLAM, and re-estimation of the map and trajectory **after** SLAM in DSG-JIT. We say "degradation", not "noise": darkness, for example, is an exposure and quantization problem, not a noise problem.

The model set below was chosen by measurement. Every candidate degradation was first located in real data (EuRoC MAV, 4Seasons), then swept, and dropped if it did not measurably hurt SLAM at realistic levels.

2. Ground rules (learned the hard way)

- **Measure real data before trusting a synthetic model.** Our synthetic darkening (Poisson-Gaussian noise) showed a large denoising win that did not exist on real dark footage. The earlier paper assumed Gaussian noise on trajectories; measured trajectory error is strongly correlated drift and bias (lag-1 autocorrelation 0.975–0.999, 60–99% of power below 0.1 Hz), not white noise.
- **Report validity before accuracy:** valid-run rate, coverage, resets/maps, start-up delay, then ATE. ATE alone hides tracking loss. ORB-SLAM3 saves only the final map, so require one map and ≥ 90% coverage of the whole sequence.
- **Statistics:** ≥ 10 runs per cell (EuRoC minimum detectable effect ≈ 5 mm at n = 10); pair arms on the same seed; Holm-correct across cells.
- **Every correction must leave clean input unchanged** and be gated on a measured quality signal.

3. Models we are keeping

Module A: Calibration and timing errors (post-SLAM; main contribution)

Injected exactly by `calib_perturb.py`; only the magnitudes are assumptions. ORB-SLAM3 estimates none of these, and the damage is **silent**: runs stay valid with no resets, so tracking health alone does not reveal it. Sweep: MH01 + MH03, 6 errors × 3 tiers × 10 seeds + clean, 380 runs. Δ = median ATE change vs clean (clean 38 / 28 mm).

Error	Tier 1 (realistic)	Tier 2 (mishandled)	Tier 3 (stress)	Priority
Stereo extrinsic rotation	0.2°: +58 / +62 mm	1.0°: +136 / +85 mm	2.5°: 5/10 runs fail	**1**. Damage follows yaw (depth bias); pitch breaks matching
Camera–IMU time offset	5 ms: +4 / +5 mm	15 ms: +10 / +18 mm	30 ms: +21 / +44 mm	**2**. IMU-early worse than IMU-late
IMU scale factor*	0.5%: +4 / +11 mm	1%: +7 / +22 mm	2%: +19 / +56 mm	**3**
Camera–IMU rotation	0.25°: +2 / +2 mm	0.8°: +4 / +9 mm	2.0°: +34 / +43 mm	secondary
IMU axis misalignment*	0.05°: 0 / +1 mm	0.5°: +9 / +11 mm	2.0°: +53 / +62 mm	secondary

Values are MH01 / MH03. * tiers are provisional (no survey source). Camera–IMU translation was benign (≤ +11 mm at 6 cm) and is dropped.

Correction: batch re-estimation of these parameters from a finished SLAM run, without ground truth, in DSG-JIT, plus a residual-based miscalibration detector. Prior work estimates each parameter individually (stereo self-calibration since 2009; time offset in VINS/OpenVINS; IMU intrinsics in OpenVINS; targetless batch calibration in Maye, Schneider, iKalibr). Our gap is the **joint, post-hoc, system-agnostic** recovery from a SLAM that does no calibration, and a detector that names the faulty parameter. See `reports/Calibration recovery literature check.md`.

Module B: Photometric / exposure (pre-SLAM)

Model, redefined: real darkness is signal compression, black clipping and 8-bit quantization, **not** Poisson-Gaussian noise. Two real regimes were measured:

- **Patchy dark** (EuRoC MH04/05): dark regions next to bright lights; 12–13% of frames at severity L3 or worse, 2–4% at L5, for ≤ 10 s at a time.
- **Uniformly crushed dark** (4Seasons `old_town_4` night): median frame mean 6.5 DN, exposure fixed at 21 ms, yet nearly noise-free (σ ≈ 0.2 DN; 65–80% of neighbouring pixels identical, about 10 grey levels in use).

4Seasons, same route	Good runs	Median ATE	Median tracking failures
Day, no correction (10 runs)	10/10	7.0 m	0
Night, no correction (C0, 30 runs)	12/30	43 m	260
Night, saturation-safe gain (C2s, 30 runs)	20/30	27 m	46 (better, p = 0.0003; composite p = 0.004)
Night, CLAHE (C5, 10 runs)	2/10	43 m	320 (no effect)
Night, lowered FAST threshold (C1, 10 runs)	0/10	–	307 (worse, p = 0.02)

"Good" = one map, ≥ 90% coverage, ATE < 1 km (route 5.7 km). On EuRoC, gain cut dark-window drift by 16–20%; NLM denoising added nothing.

Correction: saturation-safe gain gated on a brightness monitor (the only arm that helped). Because 8-bit images have already lost the information, the larger lever is likely upstream: camera exposure/gain control before quantization.

4. Models we dropped (report as measured negative results)

Candidate	Why it is out
Image denoising (NLM)	No benefit on real dark EuRoC or 4Seasons; real dark frames are not noise-limited
Motion blur	No drift cost per metre on Vicon sequences (V1_03 2.1% blurred vs 2.3% sharp; V2_03 p = 0.49); only matters when also dark
Exposure flicker / L–R mismatch	≤ 0.53 EV worst case, steps > 0.25 EV in < 2% of frames; folded into Module B
Stochastic sensor noise	IMU white noise harmless below 10× datasheet; fixed-pattern noise null at realistic levels
Gaussian trajectory noise	Measured absent: errors are correlated drift and repeatable bias
Camera–IMU translation	≤ +11 mm even at 6 cm
Rolling shutter, Bayer, JPEG, weather	Not applicable to this rig (global-shutter mono, lossless, indoor)

5. Open questions (intern tasks)

#	Question	What to do	Done when
1	**Answered:** is the 4Seasons gain effect real?	Yes at n = 30: composite p = 0.004, failures p = 0.0003, ATE of good runs p = 0.013 (valid rate alone p = 0.07)	Done
2	Can DSG-JIT recover the injected calibration errors without ground truth?	Batch re-estimation of stereo rotation, time offset, IMU scale on the existing sweep runs	Recovered parameter error, and ATE back within ~5 mm of clean
3	Can stereo yaw, accelerometer scale and time offset be told apart?	Identifiability/observability analysis under MAV and car motion	Written analysis + simulation check
4	Can a residual detector flag silent miscalibration and name the parameter?	Per-parameter residual statistics; false-alarm rate on clean runs	ROC on the 380-run sweep

#	Question	What to do	Done when
5	Does recovery work on real (not injected) miscalibration?	TUM-VI raw sequences (documented IMU scale/misalignment); UMA-VI (~5–9 ms time shift)	Recovered values match published calibration
6	Rebuild the synthetic darkening model	Scale + black clip + 8-bit quantization, noise matched to measured σ; calibrate to 4Seasons	Synthetic frames match real per-frame statistics
7	Why do C1 (low FAST) and C5 (CLAHE) fail on crushed images?	Quantization-contour hypothesis **rejected** (orb_quality.py: their corners match as well as gain-corrected ones). Both show 1–2 "bad imu" map resets per run; trace why	Cause of the resets identified
8	Firm up provisional tiers	Source IMU scale/misalignment magnitudes (datasheets, TUM-VI calibration)	Tiers cited, or sweep re-run
9	Optional: IMU bias random walk	New arm in calib_perturb.py	Swept like the other errors

6. Where things are

- Code: `dsg-jit/experiments/oasis/`. Lambda copy: `~/repos/oasis-experiments/`. Key scripts: `calib_perturb.py`, `real_correct_sequence.py`, `fourseasons_to_euroc.py`, `fourseasons_analyze.py`, `errchar.py`, `phase2_suite.py`, `run_oasis.sh`.
- Results: `dsg-jit/experiments/oasis/results/` (calibration sweep, Vicon blur, error character, 4Seasons).
- Data on lambda: `/data/tkocher/datasets/` (EuRoC, 4Seasons day + night). Keep all data on `/data`; `/home` is shared and full. Run outputs go to `/data/tkocher/oasis_results/`.
- 4Seasons note: its images are pre-rectified; the 1.48° rectifying rotation is recovered in the converter and folded into IMU.T_b_c1. Omitting it is a tier-3 camera–IMU rotation error. 4Seasons is licensed CC BY-NC-SA (non-commercial).